Dietary Mutagen Exposure and Risk of Pancreatic Cancer

To investigate the association between dietary exposure to food mutagens and risk of pancreatic cancer, we conducted a hospital-based case-control study at the University of Texas M. D. Anderson Cancer Center during June 2002 to May 2006. Atotal of 626 cases and 530 noncancer controls were frequency matched for race, sex and age (±5 years). Dietary exposure information was collected via personal interview using a meat preparation questionnaire. A significantly greater portion of the cases than controls showed a preference to well-done pork, bacon, grilled chicken, and pan-fried chicken, but not to hamburger and steak. Cases had a higher daily intake of food mutagens and mutagenicity activity (revertants per gram of daily meat intake) than controls did. The daily intakes of 2-amino-3,4,8-trimethylimidazo[4,5—f]quinoxaline (DiMeIQx) and benzo(a)pyrene (BaP), as well as the mutagenic activity, were significant predictors for pancreatic cancer ($P = 0.008$, 0.031, and 0.029, respectively) with adjustment of other confounders. A significant trend of elevated cancer risk with increasing DiMeIQx intake was observed in quintile analysis (Ptrend= 0.024). Ahigher intake of dietary mutagens (those in the two top quintiles) was associated with a 2-fold increased risk of pancreatic cancer among those without a family history of cancer but not among those with a family history of cancer. Apossible synergistic effect of dietary mutagen exposure and smoking was observed among individuals with the highest level of exposure (top 10%) to PhIP and BaP, Pinteraction= 0.09 and 0.099, respectively. These data support the hypothesis that dietary mutagen exposure alone and in interaction with other factors contribute to the development of pancreatic cancer.